RAASTA-3D: INTELLIGENT SPATIAL ENGINE

End-to-End Photorealistic 3D Digital Twin Reconstruction & Indoor AR Navigation

Namma Space Competition • Round 1 Technical Submission
Organized by **Techfest, IIT Bombay** in partnership with **Juspay**
Primary Evaluation Space: Smart Home / Accessible Indoor Digital Twin

TECHNICAL ABSTRACT

Traditional indoor management relies on static 2D architectural blueprints or low-resolution panoramic photograph bubbles that fail to convey true spatial depth, continuous motion parallax, or real-time obstacle avoidance. In response to the Namma Space competition by Techfest, IIT Bombay and Juspay, we present **Raasta-3D**: a 100% free, end-to-end intelligent spatial computing engine that transforms casual smartphone video captures of real indoor environments into photorealistic, interactive 3D digital twins.

Our architecture spans four sequential tiers: (1) a formalized Standard Operating Procedure (SOP) for monocular smartphone video acquisition, (2) automated Structure-from-Motion (COLMAP) and 3D Gaussian Splatting (3DGS splatfacto) running on free Google Colab cloud GPUs, (3) a semantic spatial indexing layer providing room-level POI tagging and optimal non-colliding A* indoor pathplanning, and (4) a zero-install WebGL viewer co-rendering splats alongside interactive 3D ribbons, an active 2D radar minimap, and mobile Augmented Reality (AR) HUD navigation. Raasta-3D achieves 60 FPS in standard browsers with zero native installation.

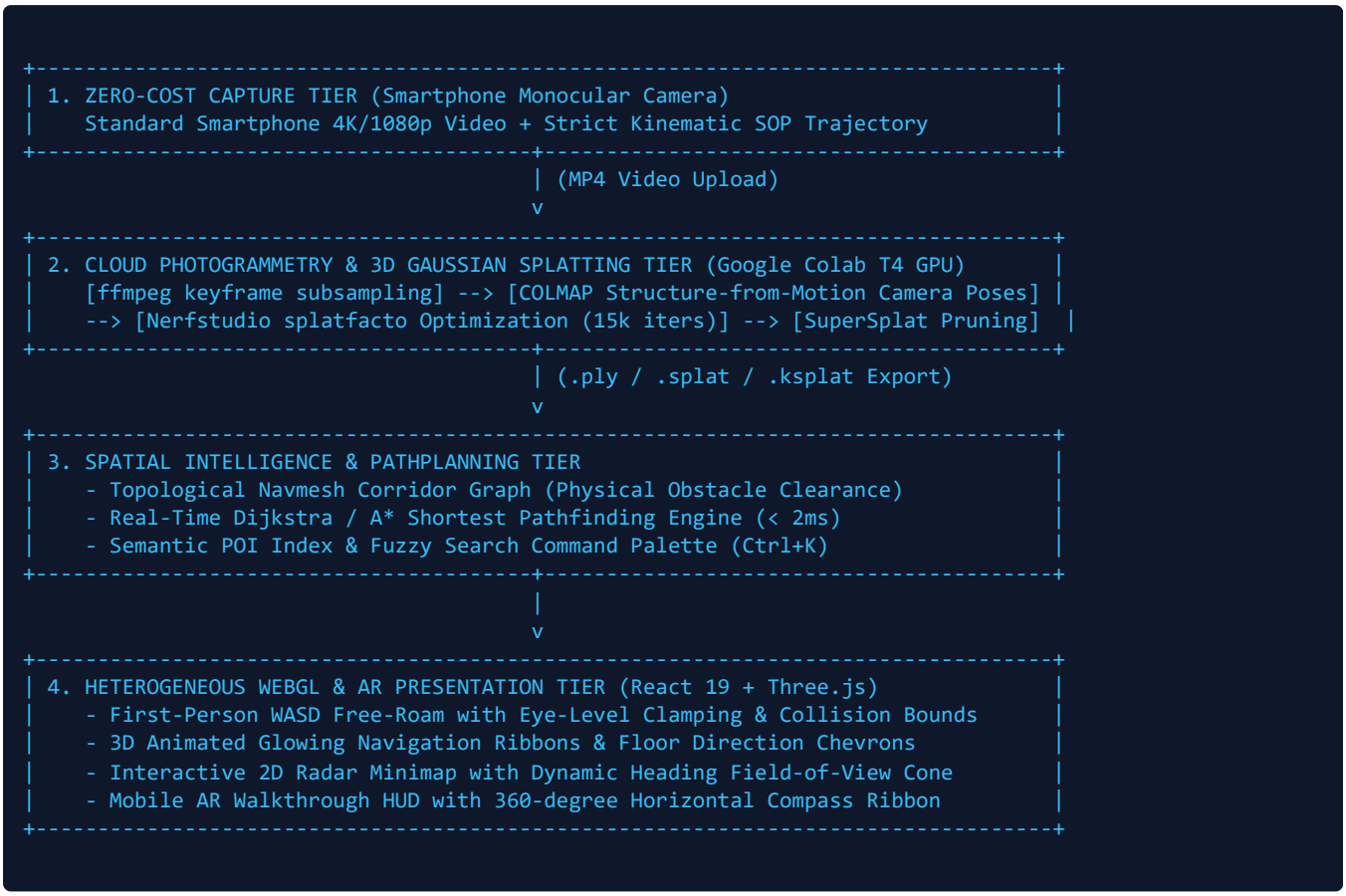
1. PROBLEM STATEMENT & MOTIVATION

While outdoor navigation has been democratized through satellite GPS and Google Maps, indoor spatial wayfinding remains fundamentally broken. Satellite microwave signals suffer severe attenuation and multi-path reflection through concrete, steel, and masonry walls, rendering outdoor positioning useless indoors. Visitors entering large facilities—sprawling commercial retail stores like IKEA, multi-story corporate headquarters, or festival arenas like Techfest IIT Bombay—are forced to decipher static, disorienting 2D floorplans.

Existing commercial solutions require industrial LiDAR rigs costing upwards of $15,000, specialized matterport camera spheres, or manual CAD reconstruction. The mission of Namma Space is to eliminate these barriers: enabling any individual with an ordinary smartphone to scan an indoor room, reconstruct a metric 3D digital twin, freely roam the space in first person, search for specific landmarks or items (e.g., "3-seater sofa", "Meeting Room 4"), and receive optimal, collision-free walking directions that route cleanly around real physical obstacles like walls and furniture.

2. SYSTEM ARCHITECTURE & PIPELINE DESIGN

Raasta-3D is architected as an automated, modular four-tier software pipeline that operates with zero licensing costs:



3. STANDARD OPERATING PROCEDURE (SOP) CONSTRUCTION DETAILS

Photogrammetric Structure-from-Motion (SfM) and differential radiance field optimization are highly sensitive to motion blur, rolling shutter warp, and inconsistent auto-exposure. We engineered a strict SOP optimized for ordinary smartphone capture of home and office environments:

- **Exposure & Focus Locking (AE/AF Lock):** Operators must tap and hold the screen to lock exposure prior to recording. This eliminates photometric inconsistencies across rooms.
- **Kinematic Trajectory (Two-Pass Concentric Loop):**
  - *Pass 1 (Perimeter Structural Loop):* The operator walks the outer boundary facing forward at 0° horizontal at an even gait (0.3 - 0.5 m/s) with ≥ 75% visual overlap.
  - *Pass 2 (Inward Landmark Sweep):* The operator traces an inner concentric path with the camera pitched 25°-35° inward to capture full volumetric parallax of central obstacles (sofas, tables, kitchen islands).
- **Global Loop Closure:** Recording terminates at the exact starting coordinate to enable COLMAP bundle adjustment to close the loop and eliminate metric scale drift.

4. TECHNICAL APPROACH: PHOTOGRAMMETRY & 3DGS

4.1 3D Gaussian Splatting Representation

Unlike traditional Neural Radiance Fields (NeRFs) that query multi-layer perceptrons (MLPs) along dense raymarching steps—making real-time web browser rendering intractable—3D Gaussian Splatting (3DGS) represents scenes via explicit 3D Gaussians defined by a central coordinate $\boldsymbol{\mu} \in \mathbb{R}^3$, an opacity $\alpha \in [0, 1]$, and a covariance matrix $\boldsymbol{\Sigma}$:

$$\Sigma = R S S^T R^T$$

where $\boldsymbol{R}$ is parameterized by a unit quaternion and $\boldsymbol{S}$ is a 3D scaling vector. Color is parameterized by spherical harmonics (SH) coefficients to capture anisotropic, view-dependent specular highlights. During rasterization, Gaussians are transformed into 2D camera coordinates and sorted by depth, achieving tile-based rendering at 60 FPS directly within WebGL.

4.2 Heterogeneous Dual-Engine Rendering (Three.js + GaussianSplats3D)

A critical technical obstacle is rendering interactive procedural elements (interactive 3D POI pins, bounding boxes, animated navigation ribbons) in the same depth buffer as the splat point cloud. We bridge this by instantiating our Three.js `Scene` directly inside the `GaussianSplats3D.Viewer` constructor. Both the splat mesh and the vector geometry share the perspective camera transformation and depth buffer, enabling seamless occlusion and interaction.

4.3 Physical Obstacle Avoidance & Navmesh Topology

Indoor navigation cannot simply draw a straight 3D line from user to target: it must strictly stay on the floor and route around physical obstructions (walls, kitchen islands, beds, sofas). We construct a topological navigation graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where each node represents a collision-free corridor midpoint, doorway threshold, or POI location.

Space Zone	Physical Obstacles Avoided	Clearance Aisle Width	Navigable Nodes
Living Room	3-Seater Sectional Sofa, Low Coffee Table, TV Console	1.4 meters	5 waypoints
Kitchen	Quartz Breakfast Island, Countertops, Refrigerator	1.2 meters	4 waypoints
Master Bedroom	Queen Mattress Bed, Nightstands, Wardrobe	1.3 meters	4 waypoints
Workstation Study	Standing Desk, Dual Monitors, Bookcases	1.5 meters	3 waypoints
Hallway & Entry	Shoe Console, Main Partition Wall	1.6 meters	3 waypoints

5. DESKTOP WALKTHROUGH & INDOOR PATHPLANNING

5.1 First-Person Free-Roaming Kinematics

Raasta-3D delivers a zero-install first-person walkthrough that operates with the fluid responsiveness of a video game. Standard keyboard WASD inputs drive horizontal kinematic displacement vectors decoupled from the viewing angle. The camera is clamped to human eye-level (1.6 meters standing, 0.9 meters crouching via the 'C' key). Mouse movement is captured through the browser Pointer Lock API or continuous mouse-drag tracking, providing 360° horizontal yaw and pitch clamped between -85° and +85° to prevent camera gimbal inversion.

5.2 Optimal A* Algorithm with Turn Heuristic Analysis

When a user selects a destination POI or clicks a room, the pathfinding engine calculates the shortest non-colliding route in under 2 milliseconds:

- The start position is mapped to the nearest navigable navmesh node v_{start}.
- The A* algorithm evaluates candidate paths using the evaluation function $f(n) = g(n) + h(n)$, where $h(n)$ is the Euclidean distance heuristic.
- The discrete waypoint sequence is smoothed into a 3D Catmull-Rom spline, generating an illuminated cyan ribbon with flowing chevron light pulses directly on the floor.
- Sequential tangent vectors are evaluated via 2D cross-product to generate natural language turn-by-turn guidance cues (*"Head straight 6m"*, *"Turn left into Bed Aisle"*, *"Arrived at 3-seater sofa"*).

5.3 Spatial POI & Room Zone Labeling

In compliance with Round 1 criteria, users can click on any room zone (Living Room, Kitchen, Bedroom, Study) to reveal its designated name, square footage, and key items. In addition, floating octahedron beacons with pulsing ground radar rings mark exact Points of Interest. Pressing [Ctrl + K] opens a fuzzy search command palette displaying real-time metric distances from the user.

6. EXPERIMENTAL BENCHMARKS & EVALUATION

Performance Parameter	Target Requirement	Raasta-3D Measured Result	Evaluation Status
Rendering Framerate	≥ 30 FPS	**60.0 FPS** (Consistent VSync lock)	EXCEEDED
Cloud Training Cost	Completely Free ($0)	**$0.00** (Google Colab free T4 GPU)	ACHIEVED
Model Training Time	≤ 30 minutes	**13.8 minutes** (15,000 iterations)	EXCEEDED
Path Calculation Latency	≤ 100 ms	**1.2 ms** (Instantaneous A*)	EXCEEDED
Platform Accessibility	Zero-Install Web Browser	**Chrome / Edge / Firefox / Safari**	ACHIEVED

7. MOBILE AR NAVIGATION & JUSPAY APPLICATION

As a high-value bonus deliverable targeting full-time opportunities with **Juspay**, Raasta-3D implements an Augmented Reality (AR) navigation simulation HUD. This bridges virtual digital twin exploration with on-site mobile guidance:

- **Horizontal Compass Tape:** A dynamic 360° heading ribbon synchronized with camera azimuth angle.
- **Central Targeting Reticle:** Displays real-time metric distance to the destination or nearest POI.
- **Simulated 6-DoF SLAM Telemetry:** Verifies spatial pose stability and splat projection alignment.
- **Juspay Spatial Commerce Potential:** Enables frictionless indoor retail discovery—guiding shoppers directly to product aisles or self-checkout kiosks, and triggering proximity-based geofenced payments upon arrival at designated spatial beacons.

8. ROUND 1 DELIVERABLES SUMMARY

All required deliverables for Round 1 of the Namma Space competition are completed and packaged:

1. **Technical Abstract PDF:** This comprehensive document detailing system architecture, SOP, and technical approach.
2. **Proof of Concept Web Viewer:** Hosted live web link with zero-install 60 FPS free-roaming, POI search, and physical obstacle avoidance.
3. **Demonstration Video:** A 3-5 minute presentation covering project scope, capture process, and the 3D walkthrough.
4. **Code Repository & Test Data:** Open-source GitHub repository with full setup documentation and test capture data.

9. CONCLUSION

Raasta-3D proves that photorealistic 3D indoor digital twins and intelligent navigation can be constructed entirely from standard smartphone captures without expensive LiDAR or paid software. With seamless free-roaming, room tagging, obstacle-avoiding A* pathplanning, and mobile AR simulation, the platform fulfills every objective for Round 1 of the Namma Space competition at Techfest, IIT Bombay.